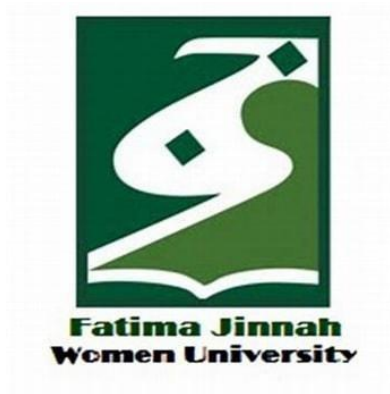


CLOUD COMPUTING

LAB NO# 06



SUBMITTED TO:

Sir Shohaib

SUBMITTED BY:

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Semester: V-A

CLOUD COMPUTING

LAB NO# 06

Task 1: Switch to root with su and back to a normal user

Step 1: Set a root password

Ubuntu disables the root account by default, so you need to assign it a password.

1. Open your terminal.
2. Run the command:

```
sudo passwd root
```

3. You will be prompted for your **current user password** (your normal account password).
4. Then it will ask you to **enter a new UNIX password**. This will be the root password for this lab.
5. Confirm the password by entering it again.

```
sudo: passwd: Command not found
amina@2023-bse-007:~$ sudo passwd root
New password:
Retype new password:
passwd: password updated successfully
amina@2023-bse-007:~$
```

Step 2: Switch to the root account

Now, use su - to switch to the root user:

1. In the terminal, type:

```
su -
```

2. Enter the root password you just set.
3. Verify you are root by checking your username and ID:

```
whoami
```

```
id
```

- whoami should return root.

- id will show uid=0(root) among other information.

```
password: password updated successfully
amina@2023-bse-007:~$ su -
Password:
root@2023-bse-007:~# whoami
root
root@2023-bse-007:~# id
uid=0(root) gid=0(root) groups=0(root)
root@2023-bse-007:~# _
```

Step 3: Switch back to your normal user

To return to your regular account:

1. Type:

exit

or press Ctrl+D.

2. Verify you're back to your normal user:

whoami

- This should return your normal username (e.g., amina).

```
root@2023-bse-007:~# exit
logout
amina@2023-bse-007:~$ whoami
amina
amina@2023-bse-007:~$ _
```

Task 2 – Create user tom and verify in passwd/group/shadow

Step 1: Create user tom

1. Open your terminal.

2. Run:

sudo adduser tom

3. The system will ask you to:

- Enter a password for tom
- Confirm the password
- Optionally enter full name, room number, etc. (you can press **Enter** to skip)

```

amina@2023-bse-007:~$ sudo adduser tom
info: Adding user `tom' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group `tom' (1001) ...
info: Adding new user `tom' (1001) with group `tom (1001)' ...
info: Creating home directory `/home/tom' ...
info: Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for tom
Enter the new value, or press ENTER for the default
  Full Name []:
  Room Number []:
  Work Phone []:
  Home Phone []:
  Other []:
Is the information correct? [Y/n]
info: Adding new user `tom' to supplemental / extra groups `users' ...
info: Adding user `tom' to group `users' ...
amina@2023-bse-007:~$

```

Step 2: Verify tom in /etc/passwd

/etc/passwd contains basic user info.

1. Run:

cat /etc/passwd

```

proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mailing List Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin
_apt:x:42:65534::/nonexistent:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
systemd-network:x:998:998:systemd Network Management:/:/usr/sbin/nologin
systemd-timesync:x:997:997:systemd Time Synchronization:/:/usr/sbin/nologin
dhcpcd:x:100:65534:DHCP Client Daemon,,,:/usr/lib/dhcpcd:/bin/false
messagebus:x:101:102::/nonexistent:/usr/sbin/nologin
systemd-resolve:x:992:992:systemd Resolver:/:/usr/sbin/nologin
pollinate:x:102:1::/var/cache/pollinate:/bin/false
polkitd:x:991:991:User for polkitd:/:/usr/sbin/nologin
syslog:x:103:104::/nonexistent:/usr/sbin/nologin
uidd:x:104:105::/run/uidd:/usr/sbin/nologin
tcpdump:x:105:107::/nonexistent:/usr/sbin/nologin
tss:x:106:108:TPM software stack,,,:/var/lib/tpm:/bin/false
landscape:x:107:109::/var/lib/landscape:/usr/sbin/nologin
fwupd-refresh:x:989:989:Firmware update daemon:/var/lib/fwupd:/usr/sbin/nologin
usbnux:x:108:46:usbnux daemon,,,:/var/lib/usbnux:/usr/sbin/nologin
amina:x:1000:1000:Amina Noor:/home/amina:/bin/bash
sshd:x:109:65534::/run/sshd:/usr/sbin/nologin
rtkit:x:110:110:RealtimeKit,,,:/proc:/usr/sbin/nologin
dnsmasq:x:999:65534:dnsmasq:/var/lib/misc:/usr/sbin/nologin
lightdm:x:111:112:Light Display Manager:/var/lib/lightdm:/bin/false
whoopsie:x:112:114::/nonexistent:/bin/false
avahi:x:113:118:Avahi mDNS daemon,,,:/run/avahi-daemon:/usr/sbin/nologin
saned:x:114:119::/var/lib/saned:/usr/sbin/nologin
colord:x:115:120:colord colour management daemon,,,:/var/lib/colord:/usr/sbin/nologin
cups-pk-helper:x:116:121:user for cups-pk-helper service,,,:/nonexistent:/usr/sbin/nologin
pulse:x:117:122:PulseAudio daemon,,,:/run/pulse:/usr/sbin/nologin
cups-browsed:x:118:121::/nonexistent:/usr/sbin/nologin
xrdp:x:119:124::/run/xrdp:/usr/sbin/nologin
tom:x:1001:1001,,,:/home/tom:/bin/bash
amina@2023-bse-007:~$

```

Step 3: Verify tom in /etc/group

/etc/group lists all groups on the system.

1. Run:

`cat /etc/group`

```
systemd-timesync:x:997:
input:x:996:
sgx:x:995:
kvm:x:994:
render:x:993:
lxd:x:101:amina
messagebus:x:102:
systemd-resolve:x:992:
_ssh:x:103:
polkitd:x:991:
crontab:x:990:
syslog:x:104:
uudd:x:105:
rdma:x:106:
tcpdump:x:107:
tss:x:108:
landscape:x:109:
fwupd-refresh:x:989:
amina:x:1000:
rtkit:x:110:
ssl-cert:x:111:
lightdm:x:112:
nopasswdlogin:x:113:
whoopsie:x:114:
netdev:x:115:
bluetooth:x:116:
scanner:x:117:saned
avahi:x:118:
saned:x:119:
colord:x:120:
lpadmin:x:121:
pulse:x:122:
pulse-access:x:123:
xrdp:x:124:
tom:x:1001:
amina@2023-bse-007:~$
```

Step 4: Verify tom in /etc/shadow

/etc/shadow stores password hashes and requires root access to view.

1. Run:

`sudo cat /etc/shadow`

```

proxy:*:20305:0:99999:7:::
www-data:*:20305:0:99999:7:::
backup:*:20305:0:99999:7:::
list:*:20305:0:99999:7:::
irc:*:20305:0:99999:7:::
_apt:*:20305:0:99999:7:::
nobody:*:20305:0:99999:7:::
systemd-networkd:!:20305:0:99999:7:::
systemd-timesyncd:!:20305:0:99999:7:::
dhcpcd:!:20305:0:99999:7:::
messagebus:!:20305:0:99999:7:::
systemd-resolve:!:20305:0:99999:7:::
pollinate:!:20305:0:99999:7:::
polkitd:!:20305:0:99999:7:::
syslog:!:20305:0:99999:7:::
uuidd:!:20305:0:99999:7:::
tcpdump:!:20305:0:99999:7:::
tss:!:20305:0:99999:7:::
landscape:!:20305:0:99999:7:::
fwupd-refresh:!:20305:0:99999:7:::
usbmux:!:20448:0:99999:7:::
amina:$6$Cg1CSQE8Tgtac/zX$3mXWnM.05.0kvxTXj0Jr066i.IK8oA4ghoT6opnynMCKYU7/0jIs.GIdB8kuk.i9FVteQLu
sshd:!:20448:0:99999:7:::
rtkit:!:20448:0:99999:7:::
dnsmasq:!:20448:0:99999:7:::
lightdm:!:20448:0:99999:7:::
whoopsie:!:20448:0:99999:7:::
avahi:!:20448:0:99999:7:::
saned:!:20448:0:99999:7:::
colord:!:20448:0:99999:7:::
cups-pk-helper:!:20448:0:99999:7:::
pulse:!:20448:0:99999:7:::
cups-browsed:!:20448:0:99999:7:::
xrdp:!:20448:0:99999:7:::
tom:$y$j9T$Pa2a.dxhEfBPbK1lI40MR0$Q/QSohKU8FoZ5cXpy3d2vc.dU2wN1z1PYZC5dtnTdk2:20463:0:99999:7:::
amina@2023-bse-007:~$

```

Task 3: Create groups; change tom's primary and secondary groups

Step 1: Create groups

Create developer, devops, and designer groups:

```
sudo groupadd developer
```

```
sudo groupadd devops
```

```
sudo groupadd designer
```

Verify they exist in /etc/group:

```
cat /etc/group
```

```
Home x Ubuntu 64-bit x
kvm:x:994:
render:x:993:
lxd:x:101:amina
messagebus:x:102:
systemd-resolve:x:992:
_ssh:x:103:
polkitd:x:991:
crontab:x:990:
syslog:x:104:
uidd:x:105:
rdma:x:106:
tcpdump:x:107:
tss:x:108:
landscape:x:109:
fwupd-refresh:x:989:
amina:x:1000:
rtkit:x:110:
ssl-cert:x:111:
lightdm:x:112:
nopasswdlogin:x:113:
whoopsie:x:114:
netdev:x:115:
bluetooth:x:116:
scanner:x:117:saned
avahi:x:118:
saned:x:119:
colord:x:120:
lpadmin:x:121:
pulse:x:122:
pulse-access:x:123:
xrdp:x:124:
tom:x:1001:
developer:x:1002:
devops:x:1003:
designer:x:1004:
amina@2023-bse-007:~$
```

Step 2: Change tom's primary group

The primary group is the group listed in /etc/passwd as the user's main group. Change it to designer:

```
sudo usermod -g designer tom
```

Verify:

```
id tom
```

```
amina@2023-bse-007:~$ sudo usermod -g designer tom
amina@2023-bse-007:~$ id tom
uid=1001(tom) gid=1004(designer) groups=1004(designer),100(users)
amina@2023-bse-007:~$
```

Step 3: Add secondary groups (developer and devops)

Add tom to multiple secondary groups **without removing existing groups**:

```
sudo usermod -aG developer,devops tom
```

Verify:

id tom

groups tom

```
amina@2023-bse-007:~$ id tom
uid=1001(tom) gid=1004(designer) groups=1004(designer),100(users)
amina@2023-bse-007:~$ groups tom
tom : designer users
amina@2023-bse-007:~$
```

Step 4: Reset secondary groups

Replace all secondary groups so only tom's own group remains:

sudo usermod -G tom tom

Verify:

id tom

groups tom

```
amina@2023-bse-007:~$ id tom
uid=1001(tom) gid=1004(designer) groups=1004(designer),100(users)
amina@2023-bse-007:~$ groups tom
tom : designer users
amina@2023-bse-007:~$
```

Task 4 – Create/delete users (Jerry, Scooby) and groups (jolly, anime)

Step 1: Create users

1. Create Jerry using adduser (interactive, creates home & password):

sudo adduser Jerry

- Follow prompts for password and optional info.

2. Create Scooby using useradd (non-interactive, no home, no password):

sudo useradd Scooby


```
Home x Ubuntu 64-bit x
Changing the user information for jerry
Enter the new value, or press ENTER for the default
  Full Name []:
  Room Number []:
  Work Phone []:
  Home Phone []:
  Other []:
Is the information correct? [Y/n]
info: Adding new user `jerry' to supplemental / extra groups `users' ...
info: Adding user `jerry' to group `users' ...
amina@2023-bse-007:~$ sudo adduser scooby
```

Step 2: Try logging in as Scooby

su - Scooby

```
amina@2023-bse-007:~$ sudo useradd scooby
amina@2023-bse-007:~$ su - scooby
Password:
^[[A^[[Asu: Authentication failure
amina@2023-bse-007:~$
```

Step 3: Set a password for Scooby

sudo passwd Scooby

```
amina@2023-bse-007:~$ sudo passwd scooby
New password:
Retype new password:
passwd: password updated successfully
amina@2023-bse-007:~$ _
```

Step 4: Try logging in as Scooby again (home directory missing)

su - Scooby

```
amina@2023-bse-007:~$ su - scooby
Password:
su: warning: cannot change directory to /home/scooby: No such file or directory
$ _
```

Step 5: Show Scooby's home directory doesn't exist

exit

cat /etc/passwd | grep Scooby

ls -ld /home/Scooby

```

$ exit
amina@2023-bse-007:~$ cat /etc/passwd | grep scooby
scooby:x:1006:1006:~/home/scooby:/bin/sh
amina@2023-bse-007:~$ ls -ld /home/scooby
ls: cannot access '/home/scooby': No such file or directory
amina@2023-bse-007:~$

```

Step 6: Create Scooby's home directory and set permissions

sudo mkdir -p /home/Scooby

sudo chown Scooby:Scooby /home/Scooby

sudo chmod 750 /home/Scooby

ls -ld /home/Scooby

```

amina@2023-bse-007:~$ sudo mkdir -p /home/scooby
amina@2023-bse-007:~$ sudo chown scooby:scooby /home/scooby
amina@2023-bse-007:~$ sudo chmod 750 /home/scooby
amina@2023-bse-007:~$ ls -ld /home/scooby
drwxr-xr-x 2 750 scooby 4096 Jan 10 07:37 /home/scooby
amina@2023-bse-007:~$ _

```

Step 7: Log in as Scooby and verify home directory

su - Scooby

pwd

ls -la

```

amina@2023-bse-007:~$ su - scooby
Password:
$ pwd
/home/scooby
$ ls -la
total 8
drwxr-xr-x 2 750 scooby 4096 Jan 10 07:37 .
drwxr-xr-x 6 root root 4096 Jan 10 07:37 ..
$ ^[S

```

Step 8: Verify users and observe Scooby's shell

exit

cat /etc/passwd | grep -E 'Jerry|Scooby'

```
$ exit
amina@2023-bse-007:~$ cat /etc/passwd | grep -E 'jerry|scooby'
jerry:x:1005:1005:::/home/jerry:/bin/bash
scooby:x:1006:1006::/home/scooby:/bin/sh
amina@2023-bse-007:~$
```

Step 9: Change Scooby's shell to /bin/bash

sudo usermod -s /bin/bash Scooby

su - Scooby

```
amina@2023-bse-007:~$ sudo usermod -s /bin/bash scooby
amina@2023-bse-007:~$ su - scooby
Password:
scooby@2023-bse-007:~$
```

Step 10: Create groups

sudo addgroup jolly # interactive

sudo groupadd anime # non-interactive

```
amina@2023-bse-007:~$ sudo addgroup jolly
[sudo] password for amina:
info: Selecting GID from range 1000 to 59999 ...
info: Adding group `jolly' (GID 1007) ...
amina@2023-bse-007:~$ sudo addgroup anime
info: Selecting GID from range 1000 to 59999 ...
info: Adding group `anime' (GID 1008) ...
```

Step 11: Verify groups

cat /etc/group | grep -E 'jolly|anime'

```
amina@2023-bse-007:~$ cat /etc/passwd | grep -E 'jolly|anime'
```

Step 12: Delete groups

sudo delgroup jolly

sudo groupdel anime

cat /etc/group | grep -E 'jolly|anime'

```
amina@2023-bse-007:~$ sudo delgroup jolly
info: Removing group `jolly' ...
amina@2023-bse-007:~$ sudo delgroup anime
info: Removing group `anime' ...
amina@2023-bse-007:~$ cat /etc/group | grep -E 'jolly|anime'
```

Step 13: Delete users

`sudo deluser --remove-home Jerry`

`sudo userdel -r Scooby`

`cat /etc/passwd | grep -E 'Jerry|Scooby'`

```
amina@2023-bse-007:~$ sudo deluser --remove-home jerry
info: Looking for files to backup/remove ...
info: Removing files ...
info: Removing crontab ...
info: Removing user `jerry' ...
amina@2023-bse-007:~$ sudo userdel -r scooby
userdel: scooby mail spool (/var/mail/scooby) not found
userdel: /home/scooby not owned by scooby, not removing
amina@2023-bse-007:~$ cat /etc/passwd | grep -E 'jolly|anime'
amina@2023-bse-007:~$
```

Task 5: Create user Student; create files; set owner/group; identify file types

Create user Student

Open terminal and run:

`sudo adduser Student`

- Set password
- Other details → press **ENTER** to skip

```

amina@2023-bse-007:~$ sudo adduser student
[sudo] password for amina:
info: Adding user `student' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group `student' (1005) ...
info: Adding new user `student' (1005) with group `student (1005)' ...
info: Creating home directory `/home/student' ...
info: Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for student
Enter the new value, or press ENTER for the default
  Full Name []:
   Room Number []:
    Work Phone []:
    Home Phone []:
      Other []:
Is the information correct? [Y/n]
info: Adding new user `student' to supplemental / extra groups `users' ...
info: Adding user `student' to group `users' ...

```

2 Switch to Student & create files

su - Student

Now create files and directory:

touch file1

mkdir -p dir1

touch dir1/file2

ls -l

```

amina@2023-bse-007:~$ su - student
Password:
student@2023-bse-007:~$ touch file1
student@2023-bse-007:~$ mkdir -p dir1
student@2023-bse-007:~$ touch dir1/file2
student@2023-bse-007:~$ ls -l
total 4
drwxrwxr-x 2 student student 4096 Jan 10 08:41 dir1
-rw-rw-r-- 1 student student   0 Jan 10 08:40 file1
student@2023-bse-007:~$

```

Change owner of file1

⚠ This must be done using **sudo**, so it works only if:

- user **tom** already exists
- group **devops** already exists

If they don't exist, create them first:

exit

sudo adduser tom

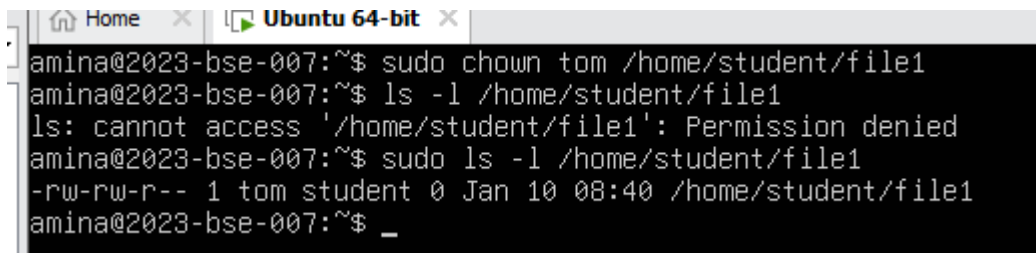
sudo addgroup devops

su - Student

Now change owner:

sudo chown tom file1

ls -l file1

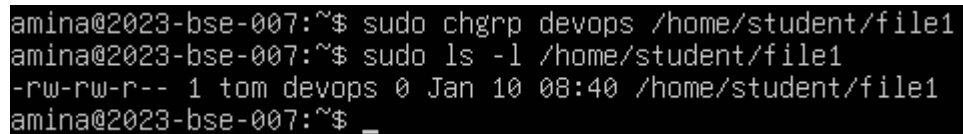
A terminal window titled 'Ubuntu 64-bit' showing a series of commands and their outputs. The user 'amina' is at a machine named '2023-bse-007'. They run 'sudo chown tom /home/student/file1', then 'ls -l /home/student/file1', which results in a 'Permission denied' error. They then run 'sudo ls -l /home/student/file1', which successfully shows the file's permissions as '-rw-rw-r--' and owner as 'tom student'.

```
amina@2023-bse-007:~$ sudo chown tom /home/student/file1
amina@2023-bse-007:~$ ls -l /home/student/file1
ls: cannot access '/home/student/file1': Permission denied
amina@2023-bse-007:~$ sudo ls -l /home/student/file1
-rw-rw-r-- 1 tom student 0 Jan 10 08:40 /home/student/file1
amina@2023-bse-007:~$ _
```

Change group of file1

sudo chgrp devops file1

ls -l file1

A terminal window showing the user 'amina' running 'sudo chgrp devops /home/student/file1'. They then run 'sudo ls -l /home/student/file1', which shows the file's permissions as '-rw-rw-r--' and group as 'devops'.

```
amina@2023-bse-007:~$ sudo chgrp devops /home/student/file1
amina@2023-bse-007:~$ sudo ls -l /home/student/file1
-rw-rw-r-- 1 tom devops 0 Jan 10 08:40 /home/student/file1
amina@2023-bse-007:~$ _
```

Identify file types & show /dev/null

Run these commands:

ls -l

ls -l dir1

ls -l /dev/null

file file1 dir1 /dev/null

```

amina@2023-bse-007:~$ sudo ls -l /home/student
total 4
drwxrwxr-x 2 student student 4096 Jan 10 08:41 dir1
-rw-rw-r-- 1 tom      devops   0 Jan 10 08:40 file1
amina@2023-bse-007:~$ sudo ls -l /home/student/dir1
total 0
-rw-rw-r-- 1 student student 0 Jan 10 08:41 file2
amina@2023-bse-007:~$ sudo ls -l /dev/null
crw-rw-rw- 1 root root 1, 3 Jan 10 07:02 /dev/null
amina@2023-bse-007:~$ sudo file /home/student/file1 /home/student/dir1 /dev/null
/home/student/file1: empty
/home/student/dir1: directory
/dev/null: character special (1/3)
amina@2023-bse-007:~$

```

Exit Student account

Exit

```

amina@2023-bse-007:~$ _

```

Task 6: Change permissions using symbolic mode

Switch to Student & verify file

su - Student

cd ~

ls -l file1

```

amina@2023-bse-007:~$ su - student
Password:
student@2023-bse-007:~$ cd ~
student@2023-bse-007:~$ ls -l file1
-rw-rw-r-- 1 student student 0 Jan 10 08:40 file1

```

Remove ALL permissions from file1

chmod a-rwx file1

ls -l file1

```

student@2023-bse-007:~$ chmod a-rwx file1
student@2023-bse-007:~$ ls -l file1
----- 1 student student 0 Jan 10 08:40 file1

```

Add READ permission to ALL

chmod +r file1

ls -l file1

```
student@2023-bse-007:~$ chmod +r file1
student@2023-bse-007:~$ ls -l file1
-r--r--r-- 1 student student 0 Jan 10 08:40 file1
```

Add EXECUTE permission to USER only

chmod u+x file1

ls -l file1

```
student@2023-bse-007:~$ chmod u+x file1
student@2023-bse-007:~$ ls -l file1
-r-xr--r-- 1 student student 0 Jan 10 08:40 file1
```

Add WRITE permission to USER and GROUP

chmod ug+w file1

ls -l file1

```
student@2023-bse-007:~$ chmod ug+w file1
student@2023-bse-007:~$ ls -l file1
-rwxrw-r-- 1 student student 0 Jan 10 08:40 file1
```

Remove ALL permissions explicitly

chmod ugo-rwx file1

ls -l file1

```
student@2023-bse-007:~$ chmod ugo-rwx file1
student@2023-bse-007:~$ ls -l file1
----- 1 student student 0 Jan 10 08:40 file1
student@2023-bse-007:~$ _
```

Task 7 – Change Permissions using SET (=) Symbolic Mode

Ensure Student context

su - Student

cd ~

ls -l file1


```

amina@2023-bse-007:~$ su - student
Password:
student@2023-bse-007:~$ cd ~
student@2023-bse-007:~$ ls -l file1
----- 1 student student 0 Jan 10 08:40 file1
student@2023-bse-007:~$ _

```

Set ALL permissions to rwx

chmod u=rwx,g=rwx,o=rwx file1

ls -l file1

```

student@2023-bse-007:~$ chmod u=rwx,g=rwx,o=rwx file1
student@2023-bse-007:~$ ls -l file1
-rwxrwxrwx 1 student student 0 Jan 10 08:40 file1
student@2023-bse-007:~$ _

```

Remove execute from GROUP and OTHERS

chmod g=rw,o=rw file1

ls -l file1

```

student@2023-bse-007:~$ chmod u=rw,g=rw,o=rw file1
student@2023-bse-007:~$ ls -l file1
-rw-rw-rw- 1 student student 0 Jan 10 08:40 file1
student@2023-bse-007:~$ _

```

Remove ALL permissions from everyone

chmod u=,g=,o= file1

ls -l file1

```

student@2023-bse-007:~$ chmod u=,g=,o= file1
student@2023-bse-007:~$ ls -l file1
----- 1 student student 0 Jan 10 08:40 file1
student@2023-bse-007:~$ _

```

Task 8 – Change Permissions using Numeric (Octal) Mode

Ensure Student context

su - Student

cd ~

ls -l file1

```
amina@2023-bse-007:~$ su - student
Password:
student@2023-bse-007:~$ cd ~
student@2023-bse-007:~$ ls -l file1
----- 1 student student 0 Jan 10 08:40 file1
student@2023-bse-007:~$ _
```

chmod 777 → full access to everyone

chmod 777 file1

ls -l file1

```
student@2023-bse-007:~$ ls l file1
ls: cannot access 'l': No such file or directory
file1
student@2023-bse-007:~$ ls -l file1
-rwxrwxrwx 1 student student 0 Jan 10 08:40 file1
student@2023-bse-007:~$
```

chmod 700 → only owner has access

chmod 700 file1

ls -l file1

```
student@2023-bse-007:~$ ls -l file1
-rwx----- 1 student student 0 Jan 10 08:40 file1
student@2023-bse-007:~$ _
```

chmod 744 → owner full, others read

chmod 744 file1

ls -l file1

```
student@2023-bse-007:~$ chmod 744 file1
student@2023-bse-007:~$ ls -l file1
-rwxr--r-- 1 student student 0 Jan 10 08:40 file1
student@2023-bse-007:~$
```

chmod 640 → owner read/write, group read

chmod 640 file1

ls -l file1

```
student@2023-bse-007:~$ chmod 640 file1
student@2023-bse-007:~$ ls -l file1
-rw-r----- 1 student student 0 Jan 10 08:40 file1
student@2023-bse-007:~$ chmod 664 file1
```

chmod 664 → owner/group rw, others read

chmod 664 file1

ls -l file1

```
student@2023-bse-007:~$ chmod 664 file1
student@2023-bse-007:~$ ls -l file1
-rw-rw-r-- 1 student student 0 Jan 10 08:40 file1
student@2023-bse-007:~$ chmod 775 file1
```

chmod 775 → owner/group full, others rx

chmod 775 file1

ls -l file1

```
student@2023-bse-007:~$ chmod 775 file1
student@2023-bse-007:~$ ls -l file1
-rwxrwxr-x 1 student student 0 Jan 10 08:40 file1
student@2023-bse-007:~$
```

chmod 750 → owner full, group rx

chmod 750 file1

ls -l file1

```
student@2023-bse-007:~$ chmod 750 file1
student@2023-bse-007:~$ ls -l file1
-rwxr-x--- 1 student student 0 Jan 10 08:40 file1
student@2023-bse-007:~$
```

Task 9 – Pipes, Pagers, Grep & Redirects

View syslog using less

sudo cat /var/log/syslog | less

- Scroll with ↑ ↓ / PageUp / PageDown
- Quit with q

```

2025-12-26T00:55:30.220400+00:00 2023-bse-007 kernel: BIOS-e820: [mem 0x000000000100000-0x0000000
2025-12-26T00:55:30.220401+00:00 2023-bse-007 kernel: BIOS-e820: [mem 0x0000000007fee0000-0x0000000
2025-12-26T00:55:30.220401+00:00 2023-bse-007 kernel: BIOS-e820: [mem 0x0000000007feff000-0x0000000
2025-12-26T00:55:30.220402+00:00 2023-bse-007 kernel: BIOS-e820: [mem 0x0000000007fff0000-0x0000000
2025-12-26T00:55:30.220403+00:00 2023-bse-007 kernel: BIOS-e820: [mem 0x00000000f0000000-0x0000000
2025-12-26T00:55:30.220403+00:00 2023-bse-007 kernel: BIOS-e820: [mem 0x00000000fec00000-0x0000000
2025-12-26T00:55:30.220404+00:00 2023-bse-007 kernel: BIOS-e820: [mem 0x00000000fee00000-0x0000000
2025-12-26T00:55:30.220411+00:00 2023-bse-007 kernel: BIOS-e820: [mem 0x00000000fffe0000-0x0000000
2025-12-26T00:55:30.220412+00:00 2023-bse-007 kernel: NX (Execute Disable) protection: active
2025-12-26T00:55:30.220413+00:00 2023-bse-007 kernel: APIC: Static calls initialized
2025-12-26T00:55:30.220413+00:00 2023-bse-007 kernel: SMBIOS 2.7 present.
2025-12-26T00:55:30.220414+00:00 2023-bse-007 kernel: DMI: VMware, Inc. VMware Virtual Platform/4
2025-12-26T00:55:30.220415+00:00 2023-bse-007 kernel: vmware: hypercall mode: 0x02
2025-12-26T00:55:30.220421+00:00 2023-bse-007 kernel: Hypervisor detected: VMware
2025-12-26T00:55:30.220422+00:00 2023-bse-007 kernel: vmware: TSC freq read from hypervisor : 249
2025-12-26T00:55:30.220423+00:00 2023-bse-007 kernel: vmware: Host bus clock speed read from hype
2025-12-26T00:55:30.220423+00:00 2023-bse-007 kernel: vmware: using clock offset of 4084524726 ns
2025-12-26T00:55:30.220424+00:00 2023-bse-007 kernel: tsc: Detected 2496.000 MHz processor
2025-12-26T00:55:30.220425+00:00 2023-bse-007 kernel: e820: update [mem 0x00000000-0x00000fff] us
2025-12-26T00:55:30.220431+00:00 2023-bse-007 kernel: e820: remove [mem 0x000a0000-0x000fffff] us
2025-12-26T00:55:30.220433+00:00 2023-bse-007 kernel: last_pfn = 0x80000 max_arch_pfn = 0x4000000
2025-12-26T00:55:30.220434+00:00 2023-bse-007 kernel: total RAM covered: 3072M
2025-12-26T00:55:30.220435+00:00 2023-bse-007 kernel: Found optimal setting for mtrr clean up
2025-12-26T00:55:30.220436+00:00 2023-bse-007 kernel: gran_size: 64K chunk_size: 64K
2025-12-26T00:55:30.220437+00:00 2023-bse-007 kernel: MTRR map: 6 entries (5 fixed + 1 variable);
2025-12-26T00:55:30.220438+00:00 2023-bse-007 kernel: x86/PAT: Configuration [0-7]: WB WC UC- U
2025-12-26T00:55:30.222460+00:00 2023-bse-007 systemd-modules-load[462]: Inserted module 'msr'
2025-12-26T00:55:30.222586+00:00 2023-bse-007 systemd-modules-load[462]: Inserted module 'dm_mult
2025-12-26T00:55:30.222611+00:00 2023-bse-007 lvm[445]: 1 logical volume(s) in volume group "ub
2025-12-26T00:55:30.222622+00:00 2023-bse-007 systemd[1]: Finished systemd-sysctl.service - Apply
2025-12-26T00:55:30.222631+00:00 2023-bse-007 systemd[1]: Finished systemd-tmpfiles-setup-dev-ear
2025-12-26T00:55:30.222642+00:00 2023-bse-007 systemd[1]: Starting systemd-journal-flush.service
2025-12-26T00:55:30.222651+00:00 2023-bse-007 multipathd[478]: multipathd v0.9.4: start up
2025-12-26T00:55:30.222659+00:00 2023-bse-007 systemd[1]: Starting systemd-sysusers.service - Cre
2025-12-26T00:55:30.222676+00:00 2023-bse-007 multipathd[478]: reconfigure: setting up paths and
amina@2023-bse-007:~$

```

View syslog using more

`sudo cat /var/log/syslog | more`

- Press **SPACE** for next page
- Press **q** to quit

```

2025-12-26T00:55:30.222824+00:00 2023-bse-007 systemd[1]: systemd-ask-password-plymouth.path - Failed because of an unmet condition check (ConditionPathExists=/run/plymouth/pid).
2025-12-26T00:55:30.222834+00:00 2023-bse-007 systemd[1]: Reached target cryptsetup.target - Local
2025-12-26T00:55:30.222844+00:00 2023-bse-007 multipath: sda: failed to get sysfs uid: No such fi
2025-12-26T00:55:30.222859+00:00 2023-bse-007 multipath: sda: failed to get sgio uid: No such fil
2025-12-26T00:55:30.222872+00:00 2023-bse-007 multipathd[478]: sda: failed to get udev uid: No de
2025-12-26T00:55:30.222880+00:00 2023-bse-007 multipathd[478]: sda: failed to get path uid
2025-12-26T00:55:30.222887+00:00 2023-bse-007 multipathd[478]: uevent trigger error
2025-12-26T00:55:30.222894+00:00 2023-bse-007 systemd[1]: Found device dev-disk-by\x2duuid-2d81et
ual_S 2.
2025-12-26T00:55:30.222901+00:00 2023-bse-007 systemd[1]: Starting systemd-fsck@dev-disk-by\x2du
ile System Check on /dev/disk/by-uuid/2d81e573-1b77-475a-89af-1f7b74bd0f2a...
2025-12-26T00:55:30.222916+00:00 2023-bse-007 systemd[1]: Started systemd-fsckd.service - File Sy
2025-12-26T00:55:30.222925+00:00 2023-bse-007 lvm[585]: PV /dev/sda3 online, VG ubuntu-vg is comp
2025-12-26T00:55:30.222933+00:00 2023-bse-007 lvm[585]: VG ubuntu-vg finished
2025-12-26T00:55:30.222943+00:00 2023-bse-007 systemd-fsck[592]: /dev/sda2: clean, 317/131072 fil
2025-12-26T00:55:30.222952+00:00 2023-bse-007 systemd[1]: Finished systemd-fsck@dev-disk-by\x2du
ile System Check on /dev/disk/by-uuid/2d81e573-1b77-475a-89af-1f7b74bd0f2a.
2025-12-26T00:55:30.222961+00:00 2023-bse-007 systemd[1]: Mounting boot.mount - /boot...
2025-12-26T00:55:30.223012+00:00 2023-bse-007 systemd[1]: Mounted boot.mount - /boot.
2025-12-26T00:55:30.223028+00:00 2023-bse-007 systemd[1]: Reached target local-fs.target - Local
2025-12-26T00:55:30.223036+00:00 2023-bse-007 systemd[1]: Listening on systemd-sysext.socket - St
2025-12-26T00:55:30.223046+00:00 2023-bse-007 systemd[1]: Starting apparmor.service - Load AppAr
2025-12-26T00:55:30.223054+00:00 2023-bse-007 systemd[1]: Starting console-setup.service - Set co
2025-12-26T00:55:30.223063+00:00 2023-bse-007 systemd[1]: Starting finalrd.service - Create fina
2025-12-26T00:55:30.223083+00:00 2023-bse-007 systemd[1]: Starting ldconfig.service - Rebuild Dyn
2025-12-26T00:55:30.223091+00:00 2023-bse-007 apparmor.systemd[598]: Restarting AppArmor
2025-12-26T00:55:30.223099+00:00 2023-bse-007 systemd[1]: Starting plymouth-read-write.service -
2025-12-26T00:55:30.223106+00:00 2023-bse-007 apparmor.systemd[598]: /lib/apparmor/apparmor.syste
2025-12-26T00:55:30.223112+00:00 2023-bse-007 apparmor.systemd[598]: Reloading AppArmor profiles
2025-12-26T00:55:30.223120+00:00 2023-bse-007 systemd[1]: Starting systemd-binfmt.service - Set u
2025-12-26T00:55:30.223137+00:00 2023-bse-007 systemd[1]: Starting systemd-tmpfiles-setup.service
2025-12-26T00:55:30.223145+00:00 2023-bse-007 systemd[1]: Starting ufw.service - Uncomplicated f
2025-12-26T00:55:30.223244+00:00 2023-bse-007 systemd[1]: Finished console-setup.service - Set co
2025-12-26T00:55:30.223261+00:00 2023-bse-007 systemd[1]: Finished finalrd.service - Create fina
amina@2023-bse-007:~$

```

Grep failures/errors + head

sudo grep -E 'fail|error' /var/log/syslog | head

```

amina@2023-bse-007:~$ sudo grep -E 'fail|error' /var/log/syslog | head
2025-12-26T00:55:30.222684+00:00 2023-bse-007 multipathd[478]: sda: failed to get udev uid: No da
2025-12-26T00:55:30.222844+00:00 2023-bse-007 multipath: sda: failed to get sysfs uid: No such fi
2025-12-26T00:55:30.222859+00:00 2023-bse-007 multipath: sda: failed to get sgio uid: No such fil
2025-12-26T00:55:30.222872+00:00 2023-bse-007 multipathd[478]: sda: failed to get udev uid: No da
2025-12-26T00:55:30.222880+00:00 2023-bse-007 multipathd[478]: sda: failed to get path uid
grep: /var/log/syslog: binary file matches
2025-12-26T00:55:30.222887+00:00 2023-bse-007 multipathd[478]: uevent trigger error
2025-12-26T00:55:30.230440+00:00 2023-bse-007 kernel: ACPI: _OSC evaluation for CPUs failed, tryi
2025-12-26T00:55:30.232093+00:00 2023-bse-007 kernel: pci 0000:00:15.3: bridge window [io size 0
2025-12-26T00:55:30.232095+00:00 2023-bse-007 kernel: pci 0000:00:15.4: bridge window [io size 0
2025-12-26T00:55:30.232096+00:00 2023-bse-007 kernel: pci 0000:00:15.5: bridge window [io size 0
amina@2023-bse-007:~$

```

Redirect output (overwrite >)

sudo grep -i systemd /var/log/syslog > ~/syslog_systemd.txt

ls -l ~/syslog_systemd.txt

```

amina@2023-bse-007:~$ sudo grep -i systemd /var/log/syslog > ~/syslog_systemd.txt
grep: /var/log/syslog: binary file matches
amina@2023-bse-007:~$ ls -l ~/syslog_systemd.txt
-rw-rw-r-- 1 amina amina 425975 Jan 10 09:31 /home/amina/syslog_systemd.txt
amina@2023-bse-007:~$

```

Append output (>>)

sudo grep -i network /var/log/syslog >> ~/syslog_systemd.txt

cat ~/syslog_systemd.txt

```
2025-12-26T07:33:14.047346+00:00 2023-bse-007 kernel: e1000: Intel(R) PRO/1000 Network Driver
2025-12-26T07:33:14.047743+00:00 2023-bse-007 kernel: e1000 0000:02:01:0 eth0: Intel(R) PRO/1000
2025-12-26T07:33:14.047820+00:00 2023-bse-007 kernel: systemd[1]: Listening on systemd-networkd.s
2025-12-26T07:33:14.253334+00:00 2023-bse-007 NetworkManager[887]: <info> [1766734394.2523] host
2025-12-26T07:33:14.253555+00:00 2023-bse-007 NetworkManager[887]: <info> [1766734394.2523] host
2025-12-26T07:33:14.259517+00:00 2023-bse-007 NetworkManager[887]: <info> [1766734394.2569] dns-
ugin=systemd-resolved
2025-12-26T07:33:14.260222+00:00 2023-bse-007 NetworkManager[887]: <info> [1766734394.2579] mana
2025-12-26T07:33:14.260302+00:00 2023-bse-007 NetworkManager[887]: <info> [1766734394.2579] mana
2025-12-26T07:33:14.275709+00:00 2023-bse-007 NetworkManager[887]: <info> [1766734394.2730] Load
orkManager/1.46.0/libnm-device-plugin-team.so)
2025-12-26T07:33:14.278030+00:00 2023-bse-007 NetworkManager[887]: <info> [1766734394.2779] Load
orkManager/1.46.0/libnm-device-plugin-wwan.so)
2025-12-26T07:33:14.285251+00:00 2023-bse-007 NetworkManager[887]: <info> [1766734394.2839] Load
workManager/1.46.0/libnm-device-plugin-bluetooth.so)
2025-12-26T07:33:14.289550+00:00 2023-bse-007 NetworkManager[887]: <info> [1766734394.2864] Load
rkManager/1.46.0/libnm-device-plugin-adsl.so)
2025-12-26T07:33:14.290958+00:00 2023-bse-007 NetworkManager[887]: <info> [1766734394.2908] Load
orkManager/1.46.0/libnm-device-plugin-wifi.so)
2025-12-26T07:33:14.294676+00:00 2023-bse-007 NetworkManager[887]: <info> [1766734394.2943] mana
e file
2025-12-26T07:33:14.297493+00:00 2023-bse-007 NetworkManager[887]: <info> [1766734394.2954] mana
file
2025-12-26T07:33:14.297692+00:00 2023-bse-007 NetworkManager[887]: <info> [1766734394.2954] mana
2025-12-26T07:33:14.298979+00:00 2023-bse-007 dbus-daemon[824]: [system] Activating via systemd:
edesktop.nm-dispatcher.service' requested by ':1.10' (uid=0 pid=887 comm="/usr/sbin/NetworkManage
2025-12-26T07:33:14.301971+00:00 2023-bse-007 NetworkManager[887]: <info> [1766734394.3008] sett
gnu/NetworkManager/1.46.0/libnm-settings-plugin-ifupdown.so")
2025-12-26T07:33:14.302139+00:00 2023-bse-007 NetworkManager[887]: <info> [1766734394.3008] sett
2025-12-26T07:33:14.302186+00:00 2023-bse-007 NetworkManager[887]: <info> [1766734394.3008] ifup
2025-12-26T07:33:14.302236+00:00 2023-bse-007 NetworkManager[887]: <info> [1766734394.3011] ifup
2025-12-26T07:33:14.316595+00:00 2023-bse-007 systemd[1]: Starting NetworkManager-dispatcher.serv
2025-12-26T07:33:14.377164+00:00 2023-bse-007 systemd[1]: Started NetworkManager-dispatcher.servi
2025-12-26T07:33:15.227145+00:00 2023-bse-007 systemd[1]: Reloading requested from client PID 103
2025-12-26T07:33:16.049019+00:00 2023-bse-007 NetworkManager[887]: <info> [1766734396.0483] dhcp
amina@2023-bse-007:~$
```

If /var/log/syslog does NOT exist (Very Common)

journalctl (alternative)

sudo journalctl | less

Filter journal logs and redirect

sudo journalctl -u systemd | grep -i error > ~/journal_errors.txt

cat ~/journal_errors.txt

```
amina@2023-bse-007:~$ sudo journalctl | less
sudo: journalctl: command not found
amina@2023-bse-007:~$ sudo journalctl -u systemd | grep -i error > ~/journal_errors.txt
amina@2023-bse-007:~$ cat ~/journal_errors.txt
amina@2023-bse-007:~$
```

Task 10: Script setup.sh – variables, command substitution, file/dir checks, permissions (use vim)

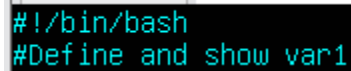
Step 1: Create script with shebang

1. Open vim:

vim setup.sh

2. Add the first line:

```
#!/bin/bash
```



```
#!/bin/bash
#Define and show var1
```

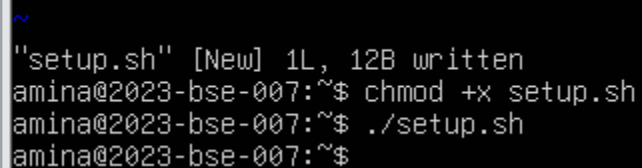
3. Save & quit :wq

4. Make executable:

```
chmod +x setup.sh
```

5. Run it:

```
./setup.sh
```



```
~
"setup.sh" [New] 1L, 12B written
amina@2023-bse-007:~$ chmod +x setup.sh
amina@2023-bse-007:~$ ./setup.sh
amina@2023-bse-007:~$
```

Step 2: Define a variable and echo it

1. Open vim to append:

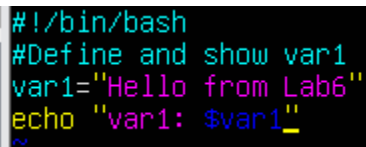
```
vim setup.sh
```

2. Add:

```
# Define and show var1
```

```
var1="Hello from Lab 6"
```

```
echo "var1: $var1"
```



```
#!/bin/bash
#Define and show var1
var1="Hello from Lab6"
echo "var1: $var1"
```

3. Save & quit :wq

4. Run:

```
./setup.sh
```

```
"setup.sh" 4L, 76B written
amina@2023-bse-007:~$ ./setup.sh
var1: Hello from Lab6
amina@2023-bse-007:~$
```

Step 3: Save ls -l output into a variable

1. Vim appends:

Save ls -l to variable and display

```
allFiles="$(ls -l)"
```

```
echo "allFiles (ls -l):"
```

```
echo "$allFiles"
```

```
#!/bin/bash
#Define and show var1
var1="Hello from Lab6"
echo "var1: $var1"
# save ls -l to variable and display
allfiles="$(ls -l)"
echo "allfiles (ls -l):"
echo "$allfiles"
~
```

2. Save & quit

3. Run:

```
./setup.sh
```



```

"setup.sh" 8L, 177B written
amina@2023-bse-007:~$ ./setup.sh
var1: Hello from Lab6
allfiles (ls -l ):
total 115676
-rw-rw-r-- 1 amina amina      312 Dec 26 02:56 answer.md
-rw-rw-r-- 1 amina amina      253 Dec 26 05:37 apt_upgrade_vs_update.md
drwxr-xr-x 2 amina amina    4096 Dec 26 06:41 Desktop
drwxr-xr-x 2 amina amina    4096 Dec 26 06:41 Documents
drwxr-xr-x 2 amina amina    4096 Dec 26 06:41 Downloads
-rw-rw-r-- 1 amina amina 117896052 Jan  6 21:33 google-chrome-stable_current_amd64.deb
-rw-rw-r-- 1 amina amina      0 Jan 10 09:35 journal_errors.txt
-rw-r--r-- 1 root  root        0 Dec 26 12:44 key
drwxrwxr-x 4 amina amina    4096 Dec 26 03:59 lab4
drwxrwxr-x 2 amina amina    4096 Jan 10 06:41 Lab5
drwxr-xr-x 2 amina amina    4096 Dec 26 06:41 Music
drwxr-xr-x 2 amina amina    4096 Dec 26 06:41 Pictures
drwxr-xr-x 2 amina amina    4096 Dec 26 06:41 Public
-rwxrwxr-x 1 amina amina     177 Jan 10 10:05 setup.sh
drwx----- 3 amina amina    4096 Dec 26 06:08 snap
-rw-rw-r-- 1 amina amina  488679 Jan 10 09:32 syslog_systemd.txt
drwxr-xr-x 2 amina amina    4096 Dec 26 06:41 Templates
drwxrwxr-t 2 amina amina    4096 Dec 26 06:41 thinclient_drives
drwxr-xr-x 2 amina amina    4096 Dec 26 06:41 Videos
amina@2023-bse-007:~$

```

Step 4: Check if a directory exists

1. Vim append:

```
# Directory check
```

```
if [ -d "dir1" ]; then
```

```
    echo "Directory dir1 exists."
```

```
else
```

```
    echo "Directory dir1 does not exist. Creating..."
```

```
    mkdir -p "dir1"
```

```
    echo "Directory dir1 created."
```

```
Fi
```

```
#!/bin/bash
#Define and show var1
var1="Hello from Lab6"
echo "var1: $var1"
# save ls -l to variable and display
allfiles="$(ls -l)"
echo "allfiles (ls -l):"
echo "$allfiles"
# directory check
if [-d "dir1"]; then
    echo "directory dir1 exists."
else
    echo "directory dir1 doesnt exist.creating ..."
    mkdir -p "dir1"
    echo "directory dir1 created"
fi
~
~
```

2. Save & quit

3. Run:

`./setup.sh`

```
'setup.sh' 16L, 352B written
amina@2023-bse-007:~$ ./setup.sh
var1: Hello from Lab6
allfiles (ls -l):
total 115676
-rw-rw-r-- 1 amina amina      312 Dec 26 02:56 answer.md
-rw-rw-r-- 1 amina amina      253 Dec 26 05:37 apt_upgrade_vs_update.md
drwxr-xr-x 2 amina amina    4096 Dec 26 06:41 Desktop
drwxr-xr-x 2 amina amina    4096 Dec 26 06:41 Documents
drwxr-xr-x 2 amina amina    4096 Dec 26 06:41 Downloads
-rw-rw-r-- 1 amina amina 117896052 Jan  6 21:33 google-chrome-stable_current_amd64.deb
-rw-rw-r-- 1 amina amina      0 Jan 10 09:35 journal_errors.txt
-rw-r--r-- 1 root  root        0 Dec 26 12:44 key
drwxrwxr-x 4 amina amina    4096 Dec 26 03:59 lab4
drwxrwxr-x 2 amina amina    4096 Jan 10 06:41 Lab5
drwxr-xr-x 2 amina amina    4096 Dec 26 06:41 Music
drwxr-xr-x 2 amina amina    4096 Dec 26 06:41 Pictures
drwxr-xr-x 2 amina amina    4096 Dec 26 06:41 Public
-rwxrwxr-x 1 amina amina     352 Jan 10 10:08 setup.sh
drwx----- 3 amina amina    4096 Dec 26 06:08 snap
-rw-rw-r-- 1 amina amina   488679 Jan 10 09:32 syslog_systemd.txt
drwxr-xr-x 2 amina amina    4096 Dec 26 06:41 Templates
drwxrwxr-t 2 amina amina    4096 Dec 26 06:41 thinclient_drives
drwxr-xr-x 2 amina amina    4096 Dec 26 06:41 Videos
./setup.sh: line 10: [-d: command not found
directory dir1 doesnt exist.creating ...
directory dir1 created
amina@2023-bse-007:~$
```

Step 5: Check if a file exists

1. Vim append:

File check

if [-f "dir1/file2"]; then

echo "file2 already exists."

```
else  
  
echo "file2 does not exist. Creating..."  
  
touch "dir1/file2"  
  
chmod a-rwx "dir1/file2"  
  
echo "file2 created."
```

Fi

```
#Define and show var1  
var1="Hello from Lab6"  
echo "var1: $var1"  
# save ls -l to variable and display  
allfiles="$(ls -l)"  
echo "allfiles (ls -l ):"  
echo "$allfiles"  
# directory check  
if [-d "dir1"]; then  
    echo "directory dir1 exists."  
else  
    echo "directory dir1 doesnt exist.creating ..."  
    mkdir -p "dir1"  
    echo "directory dir1 created"  
fi  
# file check  
if [ -f "dir1/file2"]; then  
    echo "file2 already exists"  
else  
    echo "file2 doesnt exists. creating ..."  
    touch "dir1/file2"  
    chmod a-rwx "dir1/file2"  
    echo "file2 created."  
fi
```

2. Save & quit

3. Run:

./setup.sh

```

amina@2023-bse-007:~$ ./setup.sh
var1: Hello from Lab6
allfiles (ls -l ):
total 115680
-rw-rw-r-- 1 amina amina      312 Dec 26 02:56 answer.md
-rw-rw-r-- 1 amina amina      253 Dec 26 05:37 apt_upgrade_vs_update.md
drwxr-xr-x 2 amina amina     4096 Dec 26 06:41 Desktop
drwxr-xr-x 2 amina amina     4096 Jan 10 10:08 dir1
drwxr-xr-x 2 amina amina     4096 Dec 26 06:41 Documents
drwxr-xr-x 2 amina amina     4096 Dec 26 06:41 Downloads
-rw-rw-r-- 1 amina amina 117896052 Jan  6 21:33 google-chrome-stable_current_amd64.deb
-rw-rw-r-- 1 amina amina        0 Jan 10 09:35 journal_errors.txt
-rw-r--r-- 1 root  root        0 Dec 26 12:44 key
drwxr-xr-x 4 amina amina     4096 Dec 26 03:59 lab4
drwxr-xr-x 2 amina amina     4096 Jan 10 06:41 Lab5
drwxr-xr-x 2 amina amina     4096 Dec 26 06:41 Music
drwxr-xr-x 2 amina amina     4096 Dec 26 06:41 Pictures
drwxr-xr-x 2 amina amina     4096 Dec 26 06:41 Public
-rwxrwxr-x 1 amina amina      542 Jan 10 10:15 setup.sh
drwx----- 3 amina amina     4096 Dec 26 06:08 snap
-rw-rw-r-- 1 amina amina    488679 Jan 10 09:32 syslog_systemd.txt
drwxr-xr-x 2 amina amina     4096 Dec 26 06:41 Templates
drwxrwxr-t 2 amina amina     4096 Dec 26 06:41 thinclient_drives
drwxr-xr-x 2 amina amina     4096 Dec 26 06:41 Videos
./setup.sh: line 10: [-d: command not found
directory dir1 doesnt exist.creating ...
directory dir1 created
./setup.sh: line 18: [: missing `]'
file2 doesnt exists. creating ...
file2 created.
amina@2023-bse-007:~$

```

Step 6: Check and grant user permissions

1. Vim append:

```

# Permission checks for dir1/file2 (user permissions)

f="dir1/file2"

if [ ! -r "$f" ]; then

    echo "Read permission missing; granting to user..."

    chmod u+r "$f"

fi

if [ ! -w "$f" ]; then

    echo "Write permission missing; granting to user..."

    chmod u+w "$f"

fi

if [ ! -x "$f" ]; then

    echo "Execute permission missing; granting to user..."

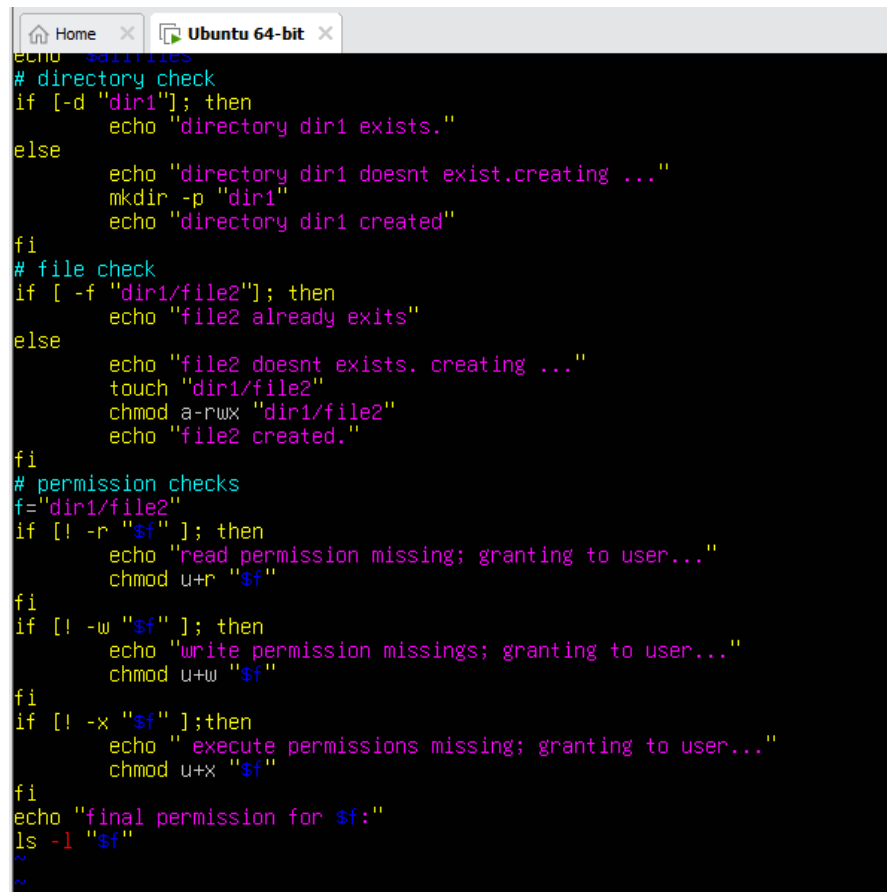
```

```
chmod u+x "$f"
```

```
fi
```

```
echo "Final permissions for $f:"
```

```
ls -l "$f"
```

A screenshot of a terminal window titled 'Ubuntu 64-bit'. The terminal displays a shell script with the following content:

```
echo $dir/files
# directory check
if [-d "$dir"]; then
    echo "directory dir1 exists."
else
    echo "directory dir1 doesnt exist.creating ..."
    mkdir -p "$dir"
    echo "directory dir1 created"
fi
# file check
if [-f "$dir/file2"]; then
    echo "file2 already exists"
else
    echo "file2 doesnt exists. creating ..."
    touch "$dir/file2"
    chmod a-rwx "$dir/file2"
    echo "file2 created."
fi
# permission checks
f="$dir/file2"
if [! -r "$f"]; then
    echo "read permission missing; granting to user..."
    chmod u+r "$f"
fi
if [! -w "$f"]; then
    echo "write permission missings; granting to user..."
    chmod u+w "$f"
fi
if [! -x "$f"]; then
    echo " execute permissions missing; granting to user..."
    chmod u+x "$f"
fi
echo "final permission for $f:"
ls -l "$f"
```

2. Save & quit

3. Run:

```
./setup.sh
```

```

amina@2023-bse-007:~$ ./setup.sh
var1: Hello from Lab6
allfiles (ls -l):
total 115680
-rw-rw-r-- 1 amina amina      312 Dec 26 02:56 answer.md
-rw-rw-r-- 1 amina amina      253 Dec 26 05:37 apt_upgrade_vs_update.md
drwxr-xr-x 2 amina amina    4096 Dec 26 06:41 Desktop
drwxrwxr-x 2 amina amina    4096 Jan 10 10:15 dir1
drwxr-xr-x 2 amina amina    4096 Dec 26 06:41 Documents
drwxr-xr-x 2 amina amina    4096 Dec 26 06:41 Downloads
-rw-rw-r-- 1 amina amina 117896052 Jan  6 21:33 google-chrome-stable_current_amd64.deb
-rw-rw-r-- 1 amina amina      0 Jan 10 09:35 journal_errors.txt
-rw-r--r-- 1 root  root        0 Dec 26 12:44 key
drwxrwxr-x 4 amina amina    4096 Dec 26 03:59 lab4
drwxrwxr-x 2 amina amina    4096 Jan 10 06:41 Lab5
drwxr-xr-x 2 amina amina    4096 Dec 26 06:41 Music
drwxr-xr-x 2 amina amina    4096 Dec 26 06:41 Pictures
drwxr-xr-x 2 amina amina    4096 Dec 26 06:41 Public
-rwxrwxr-x 1 amina amina     910 Jan 10 10:23 setup.sh
drwx----- 3 amina amina    4096 Dec 26 06:08 snap
-rw-rw-r-- 1 amina amina   488679 Jan 10 09:32 syslog_systemd.txt
drwxrwxr-x 2 amina amina    4096 Dec 26 06:41 Templates
drwxrwxr-t 2 amina amina    4096 Dec 26 06:41 thinclient_drives
drwxr-xr-x 2 amina amina    4096 Dec 26 06:41 Videos
./setup.sh: line 10: [-d: command not found
directory dir1 doesnt exist.creating ...
directory dir1 created
./setup.sh: line 18: [: missing `]'
file2 doesnt exists. creating ...
file2 created.
./setup.sh: line 28: [!: command not found
./setup.sh: line 32: [!: command not found
./setup.sh: line 36: [!: command not found
final permission for dir1/file2:
----- 1 amina amina 0 Jan 10 10:24 dir1/file2
amina@2023-bse-007:~$

```

Task 11 – Script setup.sh – argument comparisons (eq, ne, gt, lt, ge, le) and string checks

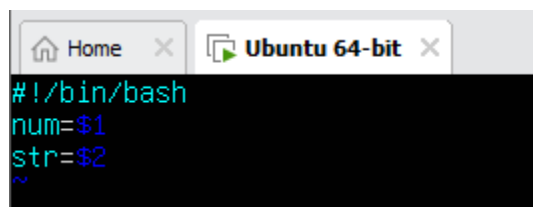
Step 0 – Create script and variables

1. Open vim setup.sh and insert:

```
#!/bin/bash
```

```
num=$1
```

```
str=$2
```



```

Home x Ubuntu 64-bit x
#!/bin/bash
num=$1
str=$2
~

```

2. Save and quit: :wq

3. Make executable astrnd run example:

`chmod +x setup.sh`

`./setup.sh 10 Student`

```
"setup.sh" 3L, 26B written
amina@2023-bse-007:~$ chmod +x setup.sh
amina@2023-bse-007:~$ ./setup.sh 10 student
amina@2023-bse-007:~$
```

Step 1 – Numeric equality (-eq)

Append to setup.sh:

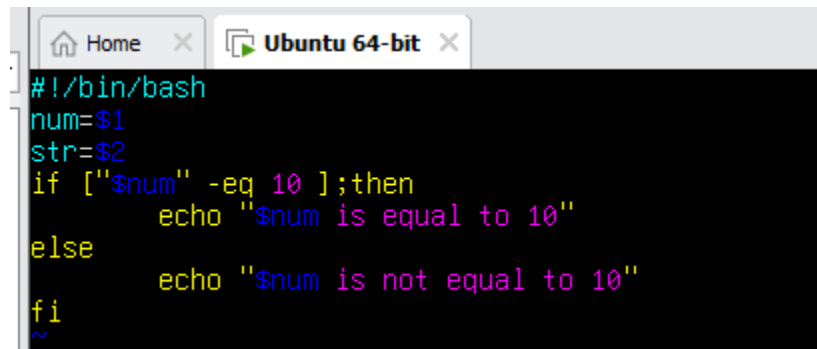
`if [ "$num" -eq 10 ]; then`

`echo "$num is equal to 10 (-eq)."`

`else`

`echo "$num is NOT equal to 10 (-eq)."`

`fi`

A screenshot of a terminal window titled 'Ubuntu 64-bit'. The terminal shows the following code being entered or displayed:

```
#!/bin/bash
num=$1
str=$2
if [ "$num" -eq 10 ];then
    echo "$num is equal to 10"
else
    echo "$num is not equal to 10"
fi
~
```

Run both examples:

`./setup.sh 10 Student`

`./setup.sh 7 Student`

```
"setup.sh" 8L, 119B written
amina@2023-bse-007:~$ ./setup.sh 10 student
./setup.sh: line 4: [10: command not found
10 is not equal to 10
amina@2023-bse-007:~$ ./setup.sh 7 student
./setup.sh: line 4: [7: command not found
7 is not equal to 10
amina@2023-bse-007:~$
```

Step 2 – Numeric not equal (-ne)

Append:

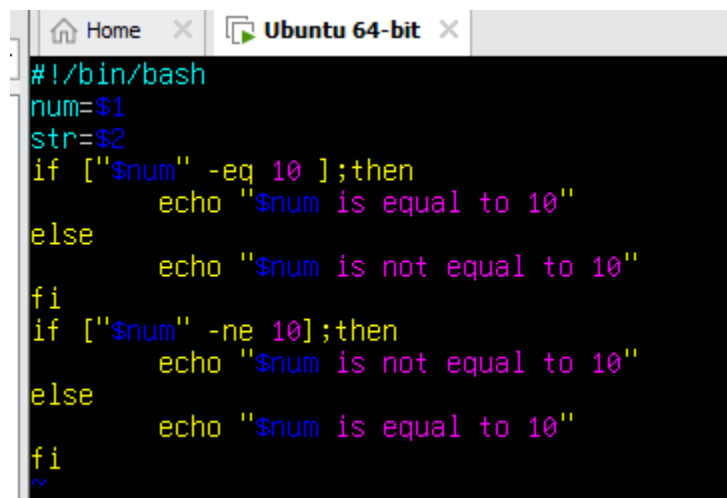
```
if [ "$num" -ne 10 ]; then
```

```
    echo "$num is not equal to 10 (-ne)."
```

```
else
```

```
    echo "$num is equal to 10 (-ne false)."
```

```
fi
```



```
#!/bin/bash
num=$1
str=$2
if [ "$num" -eq 10 ];then
    echo "$num is equal to 10"
else
    echo "$num is not equal to 10"
fi
if [ "$num" -ne 10];then
    echo "$num is not equal to 10"
else
    echo "$num is equal to 10"
fi
~
```

Run:

```
./setup.sh 7 Student
```

```
./setup.sh 10 Student
```



```

~
"setup.sh" 13L, 211B written
amina@2023-bse-007:~$ ./setup.sh 7 student
./setup.sh: line 4: [7: command not found
7 is not equal to 10
./setup.sh: line 9: [7: command not found
7 is equal to 10
amina@2023-bse-007:~$ ./setup.sh 10 student
./setup.sh: line 4: [10: command not found
10 is not equal to 10
./setup.sh: line 9: [10: command not found
10 is equal to 10
amina@2023-bse-007:~$

```

Step 3 – Greater than (-gt)

Append:

if ["\$num" -gt 10]; then

 echo "\$num is greater than 10 (-gt)."

else

 echo "\$num is NOT greater than 10 (-gt)."

fi

```

#!/bin/bash
num=$1
str=$2
if ["$num" -eq 10 ];then
    echo "$num is equal to 10"
else
    echo "$num is not equal to 10"
fi
if ["$num" -ne 10];then
    echo "$num is not equal to 10"
else
    echo "$num is equal to 10"
fi
if ["$num" -gt 10]; then
    echo "it is greater than 10"
else
    echo "it is not greater than 10"
fi
~
~

```

Run:

./setup.sh 12 Student

./setup.sh 9 Student

```

amina@2023-bse-007:~$ ./setup.sh 12 student
./setup.sh: line 4: [12: command not found
12 is not equal to 10
./setup.sh: line 9: [12: command not found
12 is equal to 10
./setup.sh: line 14: [12: command not found
it is not greater than 10
amina@2023-bse-007:~$ ./setup.sh 9 student
./setup.sh: line 4: [9: command not found
9 is not equal to 10
./setup.sh: line 9: [9: command not found
9 is equal to 10
./setup.sh: line 14: [9: command not found
it is not greater than 10
amina@2023-bse-007:~$

```

Step 4 – Less than (-lt)

Append:

```
if [ "$num" -lt 10 ]; then
```

```
    echo "$num is less than 10 (-lt)."
```

```
else
```

```
    echo "$num is NOT less than 10 (-lt)."
```

```
fi
```

```

#!/bin/bash
num=$1
str=$2
if [ "$num" -eq 10 ];then
    echo "$num is equal to 10"
else
    echo "$num is not equal to 10"
fi
if [ "$num" -ne 10];then
    echo "$num is not equal to 10"
else
    echo "$num is equal to 10"
fi
if [ "$num" -gt 10]; then
    echo "it is greater than 10"
else
    echo "it is not greater than 10"
fi
if [ "$num" -lt 10]; then
    echo "it is less than 10"
else
    echo "it is not less than 10"
fi_

```

Run:

```
./setup.sh 9 Student
```

```
./setup.sh 12 Student
```

```
amina@2023-bse-007:~$ ./setup.sh 9 student
./setup.sh: line 4: [9: command not found
9 is not equal to 10
./setup.sh: line 9: [9: command not found
9 is equal to 10
./setup.sh: line 14: [9: command not found
it is not greater than 10
./setup.sh: line 19: [9: command not found
./setup.sh: line 22: echoit is not less than 10: command not found
amina@2023-bse-007:~$ ./setup.sh 12 student
./setup.sh: line 4: [12: command not found
12 is not equal to 10
./setup.sh: line 9: [12: command not found
12 is equal to 10
./setup.sh: line 14: [12: command not found
it is not greater than 10
./setup.sh: line 19: [12: command not found
./setup.sh: line 22: echoit is not less than 10: command not found
amina@2023-bse-007:~$
```

Step 5 – Greater than or equal (-ge)

Append:

if ["\$num" -ge 10]; then

 echo "\$num is greater than or equal to 10 (-ge)."

else

 echo "\$num is NOT greater than or equal to 10 (-ge)."

fi

```
#!/bin/bash
num=$1
str=$2
if ["$num" -eq 10 ];then
    echo "$num is equal to 10"
else
    echo "$num is not equal to 10"
fi
if ["$num" -ne 10];then
    echo "$num is not equal to 10"
else
    echo "$num is equal to 10"
fi
if ["$num" -gt 10]; then
    echo "it is greater than 10"
else
    echo "it is not greater than 10"
fi
if ["$num" -lt 10]; then
    echo "it is less than 10"
else
    echo "it is not less than 10"
fi
if ["$num" -ge 10]; then
    echo "greater than or equal to 10"
else
    echo "not greater than or equal to 10"
fi_
~
^
```

Run:

./setup.sh 10 Student

./setup.sh 8 Student

```
"setup.sh" 28L, 507B written
amina@2023-bse-007:~$ ./setup.sh 10 student
./setup.sh: line 4: [10: command not found
10 is not equal to 10
./setup.sh: line 9: [10: command not found
10 is equal to 10
./setup.sh: line 14: [10: command not found
it is not greater than 10
./setup.sh: line 19: [10: command not found
./setup.sh: line 22: echoit is not less than 10: command not found
./setup.sh: line 24: [10: command not found
not greater than or equal to 10
amina@2023-bse-007:~$ ./setup.sh 8 student
./setup.sh: line 4: [8: command not found
8 is not equal to 10
./setup.sh: line 9: [8: command not found
8 is equal to 10
./setup.sh: line 14: [8: command not found
it is not greater than 10
./setup.sh: line 19: [8: command not found
./setup.sh: line 22: echoit is not less than 10: command not found
./setup.sh: line 24: [8: command not found
not greater than or equal to 10
amina@2023-bse-007:~$
```

Step 6 – Less than or equal (-le)

Append:

```
if [ "$num" -le 10 ]; then
    echo "$num is less than or equal to 10 (-le)."
else
    echo "$num is NOT less than or equal to 10 (-le)."
fi
```

```
if [ "$num" -le 10 ]; then
    echo "less than or equal to 10"
else
    echo "not less than or equal to 10"
fi
~
~
```

Run:

./setup.sh 10 Student

./setup.sh 12 Student

```
amina@2023-bse-007:~$ ./setup.sh 10 student
./setup.sh: line 4: [10: command not found
10 is not equal to 10
./setup.sh: line 9: [10: command not found
10 is equal to 10
./setup.sh: line 14: [10: command not found
it is not greater than 10
./setup.sh: line 19: [10: command not found
./setup.sh: line 22: echoit is not less than 10: command not found
./setup.sh: line 24: [10: command not found
not less than or equal to 10
amina@2023-bse-007:~$ ./setup.sh 12 student
./setup.sh: line 4: [12: command not found
12 is not equal to 10
./setup.sh: line 9: [12: command not found
12 is equal to 10
./setup.sh: line 14: [12: command not found
it is not greater than 10
./setup.sh: line 19: [12: command not found
./setup.sh: line 22: echoit is not less than 10: command not found
./setup.sh: line 24: [12: command not found
not less than or equal to 10
amina@2023-bse-007:~$
```

Step 7 – String equality (=)

Append:

```
if [ "$str" = "Student" ]; then
    echo "Second argument equals 'Student' ( = )."
```

else

echo "Second argument does NOT equal 'Student' (=)."

fi

```
if ["$str" = "student"]; then
    echo "eaul to student"
else
    echo "not equal to student"
fi
```

Run:

./setup.sh 10 Student

./setup.sh 10 Test

```
./setup.sh" 29L, 491B written
amina@2023-bse-007:~$ ./setup.sh 10 student
./setup.sh: line 4: [10: command not found
10 is not equal to 10
./setup.sh: line 9: [10: command not found
10 is equal to 10
./setup.sh: line 14: [10: command not found
it is not greater than 10
./setup.sh: line 19: [10: command not found
./setup.sh: line 22: echoit is not less than 10: command not found
./setup.sh: line 24: [student: command not found
not equal to student
amina@2023-bse-007:~$ ./setup.sh 10 test
./setup.sh: line 4: [10: command not found
10 is not equal to 10
./setup.sh: line 9: [10: command not found
10 is equal to 10
./setup.sh: line 14: [10: command not found
it is not greater than 10
./setup.sh: line 19: [10: command not found
./setup.sh: line 22: echoit is not less than 10: command not found
./setup.sh: line 24: [test: command not found
not equal to student
amina@2023-bse-007:~$
```

Step 8 – String inequality (!=)

Append:

if ["\$str" != "Student"]; then

echo "Second argument is not equal to 'Student' (!=)."

else

echo "Second argument equals 'Student' (!= false)."

fi

```

if ["$str" != "student"]; then
    echo "not equal to student"
else
    echo "equal to student"
fi

```

Run:

./setup.sh 10 Test

./setup.sh 10 Student

```

"setup.sh" 29L, 492B written
amina@2023-bse-007:~$ ./setup.sh 10 test
./setup.sh: line 4: [10: command not found
10 is not equal to 10
./setup.sh: line 9: [10: command not found
10 is equal to 10
./setup.sh: line 14: [10: command not found
it is not greater than 10
./setup.sh: line 19: [10: command not found
./setup.sh: line 22: echoit is not less than 10: command not found
./setup.sh: line 24: [test: command not found
equal to student
amina@2023-bse-007:~$ ./setup.sh 10 student
./setup.sh: line 4: [10: command not found
10 is not equal to 10
./setup.sh: line 9: [10: command not found
10 is equal to 10
./setup.sh: line 14: [10: command not found
it is not greater than 10
./setup.sh: line 19: [10: command not found
./setup.sh: line 22: echoit is not less than 10: command not found
./setup.sh: line 24: [student: command not found
equal to student
amina@2023-bse-007:~$

```

Step 9 – Check if second argument is empty (-z)

Append:

```
if [ -z "$str" ]; then
```

```
    echo "Second argument is empty (zero-length)."
```

```
else
```

```
    echo "Second argument is not empty."
```

```
fi
```

```
if [-z "$str" ]; then
    echo "empty"
else
    echo "not empty"
fi
```

Run:

`./setup.sh 10`

`./setup.sh 10 Student`

```
"setup.sh" 29L, 461B written
amina@2023-bse-007:~$ ./setup.sh 10
./setup.sh: line 4: [10: command not found
10 is not equal to 10
./setup.sh: line 9: [10: command not found
10 is equal to 10
./setup.sh: line 14: [10: command not found
it is not greater than 10
./setup.sh: line 19: [10: command not found
./setup.sh: line 22: echoit is not less than 10: command not found
./setup.sh: line 24: [-z: command not found
not empty
amina@2023-bse-007:~$ ./setup.sh 10 student
./setup.sh: line 4: [10: command not found
10 is not equal to 10
./setup.sh: line 9: [10: command not found
10 is equal to 10
./setup.sh: line 14: [10: command not found
it is not greater than 10
./setup.sh: line 19: [10: command not found
./setup.sh: line 22: echoit is not less than 10: command not found
./setup.sh: line 24: [-z: command not found
not empty
amina@2023-bse-007:~$
```

Task 12 – Script setup.sh – print all arguments with a for loop

Step 1 – Create script with shebang and comment

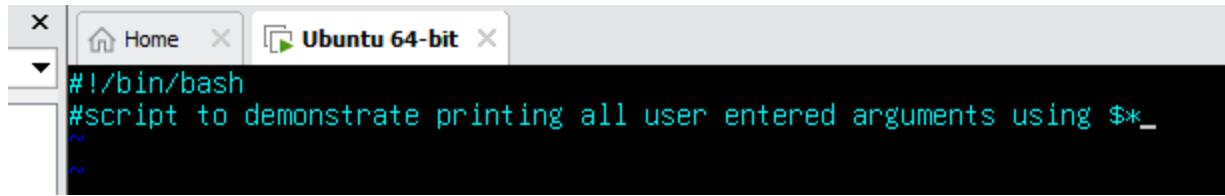
1. Open vim and overwrite setup.sh:

`vim setup.sh`

2. Insert the following lines:

`#!/bin/bash`

`# Script to demonstrate printing all user-entered arguments using $*`

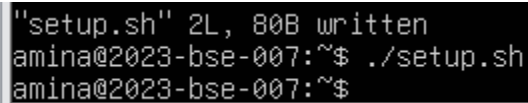
A terminal window titled 'Ubuntu 64-bit' showing a script. The script starts with a shebang line '#!/bin/bash' and a comment '#script to demonstrate printing all user entered arguments using \$*_'. There are two tilde '~' characters on the following lines.

```
#!/bin/bash
#script to demonstrate printing all user entered arguments using $*_
~
~
```

3. Save and quit: :wq

4. Run the script (no arguments expected yet):

`./setup.sh`

A terminal window showing the output of running the script. It displays the file size and permissions, followed by the command being executed and the prompt.

```
"setup.sh" 2L, 80B written
amina@2023-bse-007:~$ ./setup.sh
amina@2023-bse-007:~$
```

Step 2 – Append for loop to print all arguments

1. Re-open setup.sh in vim:

`vim setup.sh`

2. Append the following lines **below the comment**:

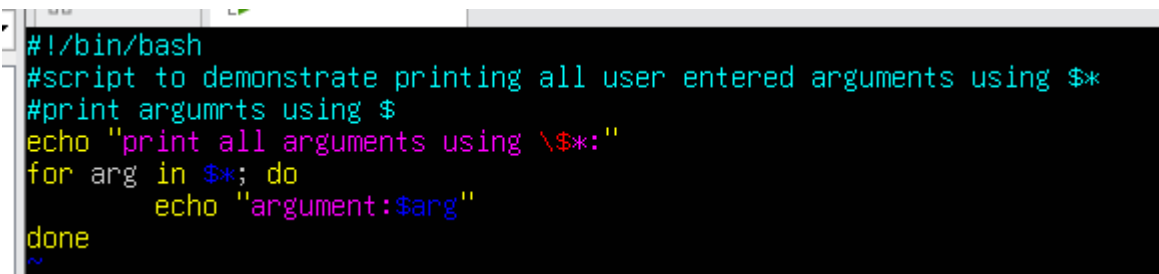
`# Print all arguments using $*`

`echo "Printing all arguments using \${*}:"`

`for arg in $*; do`

`echo "Argument: $arg"`

`done`

A terminal window showing the updated script. It includes the previous comment and adds a new comment, an echo statement, a for loop, and a done statement.

```
#!/bin/bash
#script to demonstrate printing all user entered arguments using $*_
#print argumrts using $
echo "print all arguments using \${*}:"
for arg in $*; do
    echo "argument:$arg"
done
~
```

3. Save and quit: :wq

4. Make the script executable (if not already):

`chmod +x setup.sh`

5. Run the script with example arguments:

`./setup.sh one "two words" three`

```
"setup.sh" 7L, 187B written
amina@2023-bse-007:~$ chmod +x setup.sh
amina@2023-bse-007:~$ ./setup.sh one "two words" three
print all arguments using $*:
argument:one
argument:two
argument:words
argument:three
amina@2023-bse-007:~$
```

Task 13 – Script setup.sh – while loop summation and functions

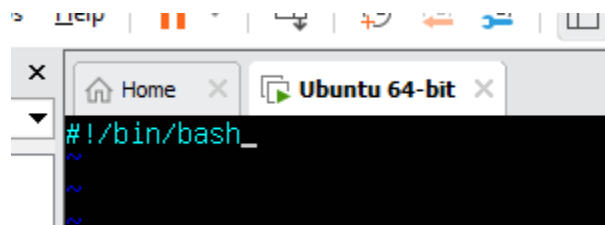
Step 1 – Create script with shebang

1. Open vim and overwrite setup.sh:

`vim setup.sh`

2. Insert:

`#!/bin/bash`



3. Save and quit: `:wq`
4. Make executable and run (no output expected):

`chmod +x setup.sh`

`./setup.sh`

```
"setup.sh" 1L, 12B written
amina@2023-bse-007:~$ chmod +x setup.sh
amina@2023-bse-007:~$ ./setup.sh
amina@2023-bse-007:~$
```

Step 2 – Add interactive while-loop summation

1. Re-open setup.sh in vim:

```
vim setup.sh
```

2. Append the following:

```
# While-loop summation (interactive)
```

```
sum=0
```

```
while true; do
```

```
    read -p "Enter a number (or 'q' to quit): " input
```

```
    if [ "$input" = "q" ]; then
```

```
        break
```

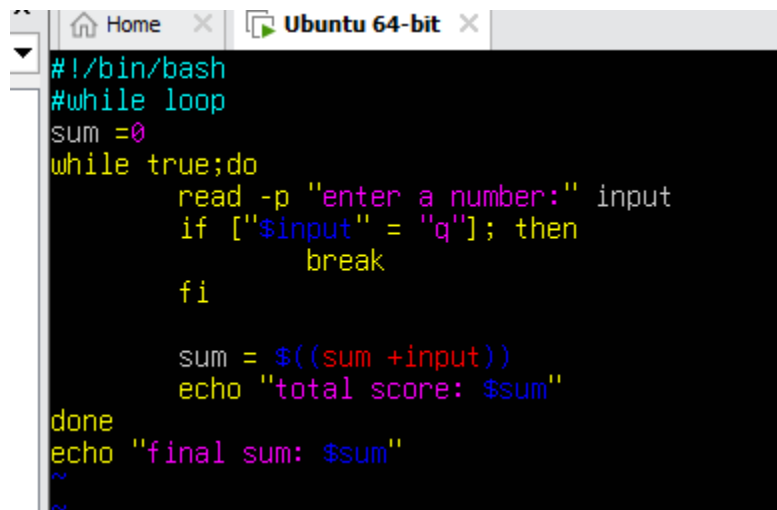
```
    fi
```

```
    sum=$((sum + input))
```

```
    echo "Total Score: $sum"
```

```
done
```

```
echo "Final total: $sum"
```

A screenshot of a terminal window titled 'Ubuntu 64-bit'. The terminal shows the following code:

```
#!/bin/bash
#while loop
sum =0
while true;do
    read -p "enter a number:" input
    if ["$input" = "q"]; then
        break
    fi

    sum = $((sum +input))
    echo "total score: $sum"
done
echo "final sum: $sum"
~
~
```

3. Save and quit: :wq

4. Run the script and demonstrate a short session:

```
./setup.sh
```

```
Enter a number (or 'q' to quit): 6
Total Score: 6
Enter a number (or 'q' to quit): 7
Total Score: 13
Enter a number (or 'q' to quit): 8
Total Score: 21
Enter a number (or 'q' to quit): 9
Total Score: 30
Enter a number (or 'q' to quit): 10
Total Score: 40
Enter a number (or 'q' to quit): q
Final total: 40
```

Step 3 – Move interactive while-loop into a function sum_two()

1. Re-open setup.sh in vim.
2. Replace/delete the standalone while-loop if you want the function to handle **input only**, then append:

```
# Function to accumulate scores interactively

sum_two() {
    sum=0
    while true; do
        read -p "Enter a number (or 'q' to quit): " input
        if [ "$input" = "q" ]; then
            break
        fi
        sum=$((sum + input))
        echo "Total Score: $sum"
    done
    echo "Function final total: $sum"
}

# Demonstrate the function

echo "Now calling sum_two function:"

sum_two
```

```
# Function to accumulate scores interactively
sum_two() {
    sum=0
    while true; do
        read -p "Enter a number (or 'q' to quit): " input
        if [ "$input" = "q" ]; then
            break
        fi
        sum=$((sum + input))
        echo "Total Score: $sum"
    done
    echo "Function final total: $sum"
}

# Demonstrate the function
echo "Now calling sum_two function:"
sum_two
```

3. Save and quit: :wq

4. Run the script and demonstrate short session for function:

./setup.sh

```
Now calling sum_two function:
Enter a number (or 'q' to quit): 3
Total Score: 3
Enter a number (or 'q' to quit): 4
Total Score: 7
Enter a number (or 'q' to quit): 5
Total Score: 12
Enter a number (or 'q' to quit): q
Function final total: 12
```

Step 4 – Add function that sums two numeric arguments and returns result

1. Re-open setup.sh in vim.

2. Append the following:

Function that sums two arguments and returns the result

```
sum_args() {
```

```
    a=$1
```

```
    b=$2
```

```
    return $((a + b))
```

```
}
```

Demonstrate sum_args function

```
echo "Now demonstrating sum_args function:"
```

```
sum_args 3 4
```

```
result=$?
```

```
echo "sum_args(3,4) returned: $result"
```

```
# Function that sums two arguments and returns the result
sum_args() {
  a=$1
  b=$2
  return $((a + b))
}

# Demonstrate sum_args function
echo "Now demonstrating sum_args function:"
sum_args 3 4
result=$?
echo "sum_args(3,4) returned: $result"
```

3. Save and quit: :wq

4. Run the script and capture the demonstration output:

```
chmod +x setup.sh
```

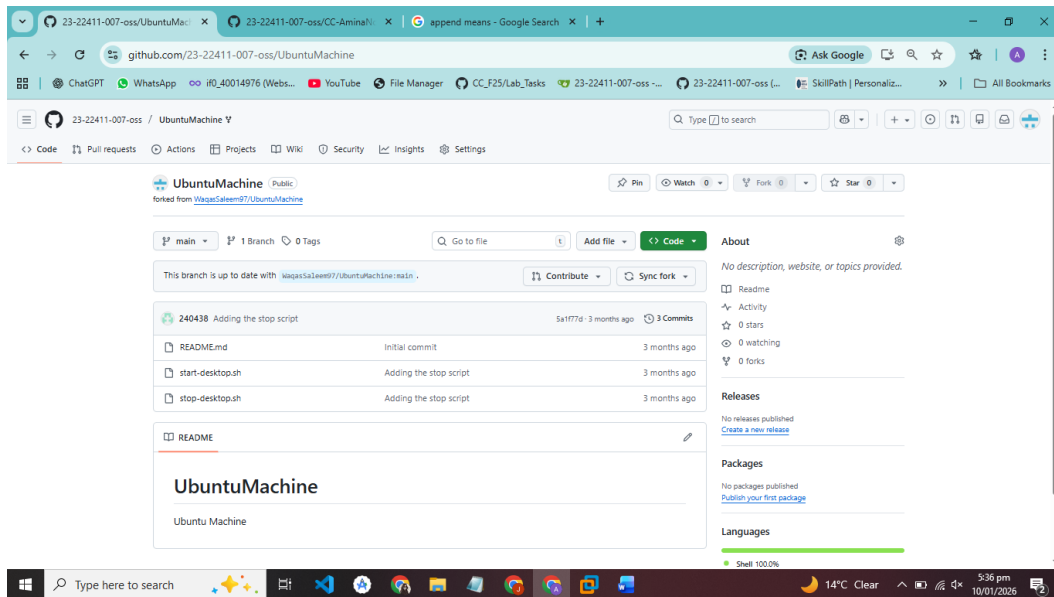
```
./setup.sh
```

```
Now demonstrating sum_args function:
sum_args(3,4) returned: 7
```

Task 14 – Codespaces GUI — fork repo, run start-desktop.sh, open VNC, stop GUI

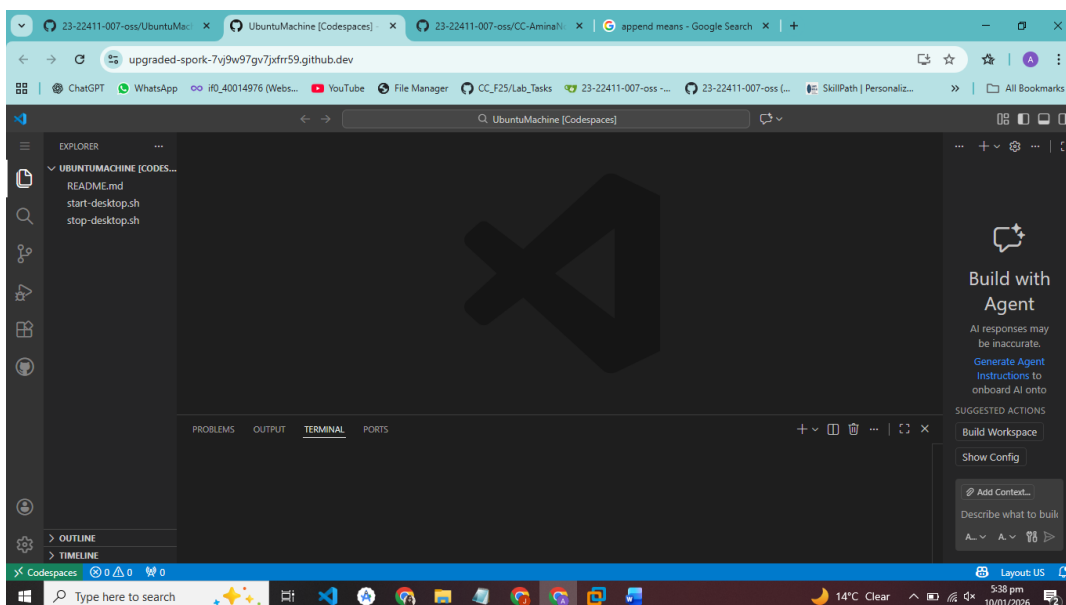
Step 1: Fork the repository to your GitHub account

1. Open the GitHub repo URL in your browser (the repo your instructor gave you).
2. Click the “Fork” button on the top-right corner.
3. Choose your GitHub account as the destination.
4. Wait for the fork to complete.



Step 2: Open a Codespace on your fork

1. Go to your forked repo on GitHub.
2. Click the green “Code” button → select Open with Codespaces → Create codespace on main (or whatever branch).
3. Wait for Codespaces to fully load.



Step 3: Verify the start/stop scripts exist and are executable

1. Open the Codespace terminal.

2. Check for the scripts:

```
ls -l start-desktop.sh stop-desktop.sh
```

3. If they are not executable, make them executable:

```
chmod +x start-desktop.sh stop-desktop.sh
```

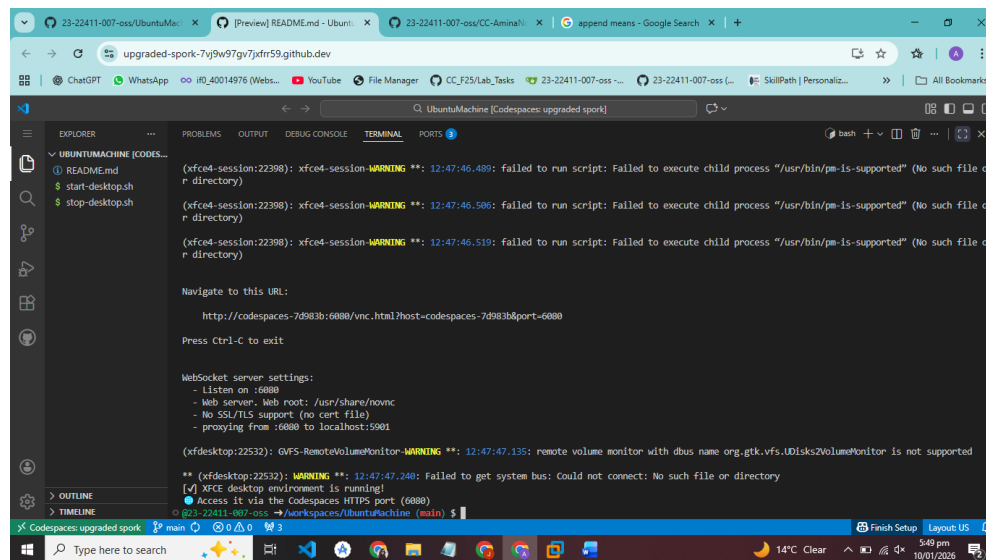
```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
@23-22411-007-oss →/workspaces/UbuntuMachine (main) $ ls -l start-dekstop.sh stop-desktop.sh
ls: cannot access 'start-dekstop.sh': No such file or directory
-rwxrwxrwx 1 codespace root 428 Jan 10 12:37 stop-desktop.sh
@23-22411-007-oss →/workspaces/UbuntuMachine (main) $ ls -l start-desktop.sh stop-desktop.sh
-rwxrwxrwx 1 codespace root 1333 Jan 10 12:37 start-desktop.sh
-rwxrwxrwx 1 codespace root 428 Jan 10 12:37 stop-desktop.sh
@23-22411-007-oss →/workspaces/UbuntuMachine (main) $ chmod +x start-desktop.sh stop-desktop.sh
@23-22411-007-oss →/workspaces/UbuntuMachine (main) $
```

Step 4: Start the desktop GUI

1. In the Codespace terminal, ensure the script is executable and run it:

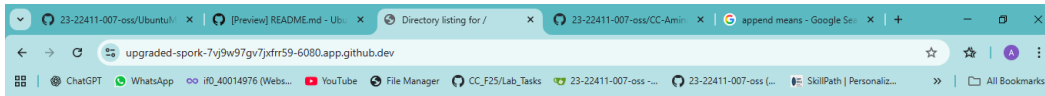
```
./start-desktop.sh
```

2. Wait for the script to finish; it should display messages like “VNC server started on port 6080”.



Step 5: Verify forwarded ports

1. In Codespaces, open the Ports view (usually bottom panel or via the sidebar).
2. Confirm port 6080 is listed and forwarded.



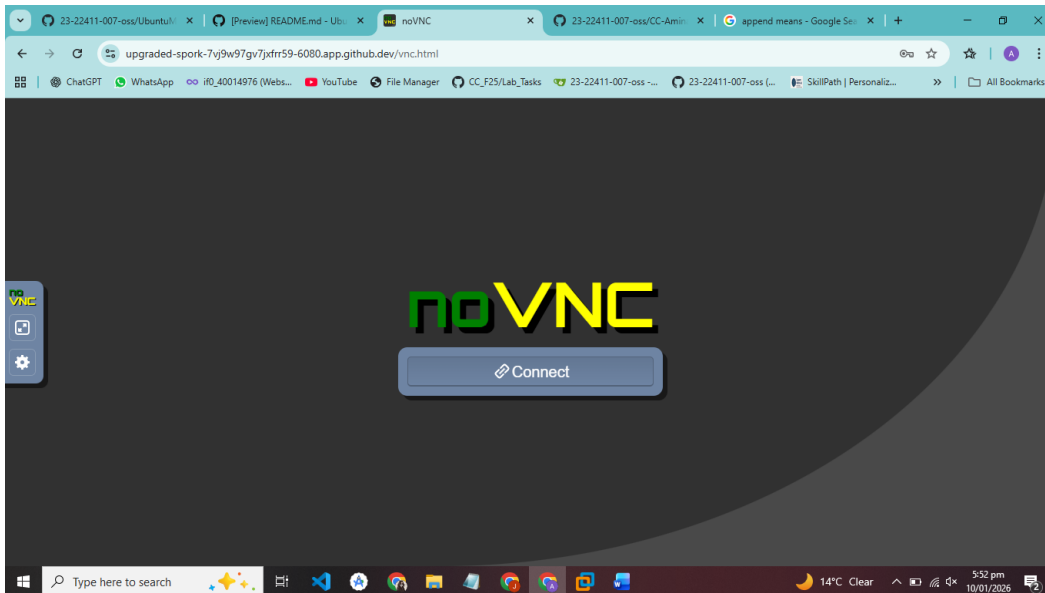
Directory listing for /

- [app/](#)
- [core/](#)
- [include/](#)
- [utils/](#)
- [vendor/](#)
- [vnc.html](#)
- [vnc_auto.html](#)
- [vnc_lite.html](#)

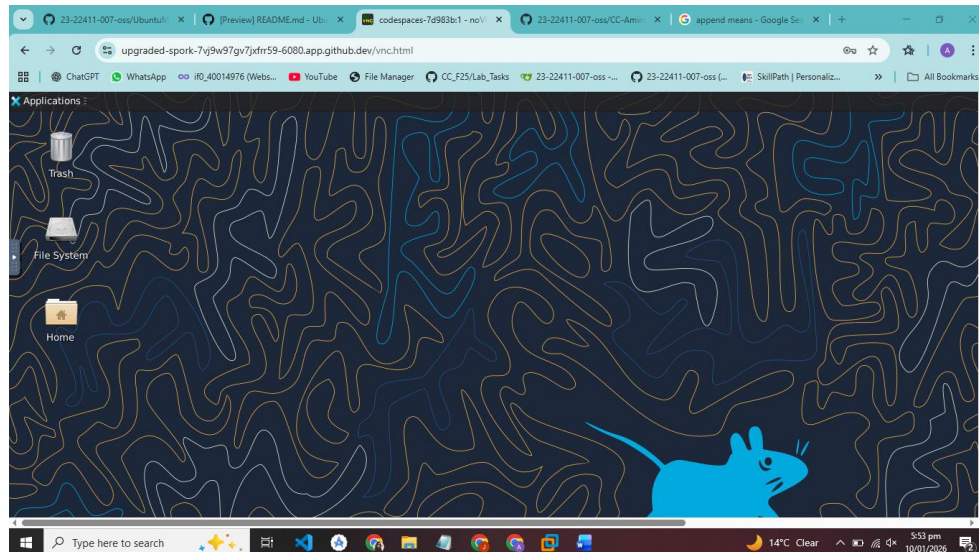
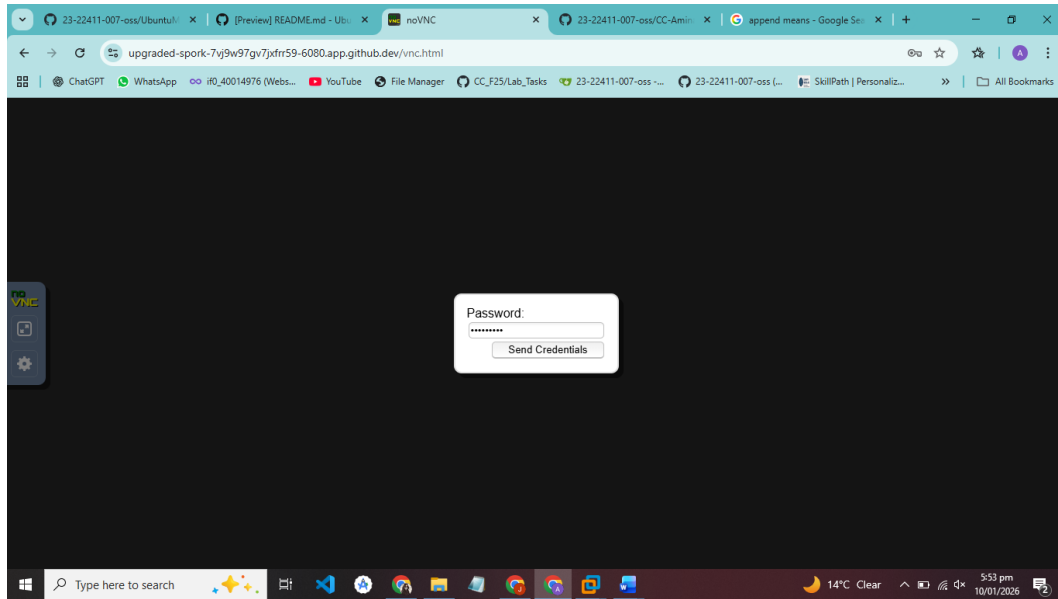


Step 6: Open the VNC HTML page

1. Click the preview URL for port 6080 or copy it into your browser.
2. Open vnc.html at that URL.



3. When prompted for a password, enter: codespace

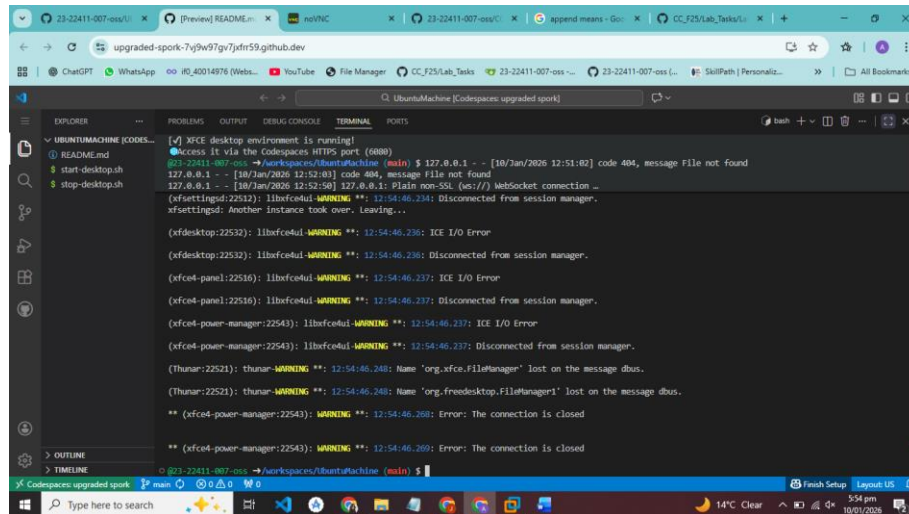


Step 7: Stop the GUI

1. Go back to Codespace terminal.
2. Run:

3. `./stop-desktop.sh`

4. Wait for the script to finish; it should show messages about stopping the VNC/GUI.



```
[*] Xfce desktop environment is running!
Access it via the CodeSpaces HTTPS port (4080)
23-22411-007-oss -> /workspaces/UbuntuMachine (min) $ 127.0.0.1 - [10/Jan/2026 12:51:02] code 404, message File not found
127.0.0.1 - [10/Jan/2026 12:52:01] code 404, message File not found
127.0.0.1 - [10/Jan/2026 12:52:50] 127.0.0.1: Plain non-SSL (ws://) Websocket connection _
(xfsettings:22512): libxfce4ui-WARNING **: 12:54:46.234: Disconnected from session manager.
xfsettings: Another instance took over. Leaving...

(xfdesktop:22532): libxfce4ui-WARNING **: 12:54:46.236: ICE I/O Error

(xfdesktop:22532): libxfce4ui-WARNING **: 12:54:46.236: Disconnected from session manager.

(xfce4-panel:22516): libxfce4ui-WARNING **: 12:54:46.237: ICE I/O Error

(xfce4-panel:22516): libxfce4ui-WARNING **: 12:54:46.237: Disconnected from session manager.

(xfce4-power-manager:22543): libxfce4ui-WARNING **: 12:54:46.237: ICE I/O Error

(xfce4-power-manager:22543): libxfce4ui-WARNING **: 12:54:46.237: Disconnected from session manager.

(thunar:22521): thunar-WARNING **: 12:54:46.248: Name 'org.xfce.FileManager' lost on the message dbus.

(thunar:22521): thunar-WARNING **: 12:54:46.248: Name 'org.freedesktop.FileManager1' lost on the message dbus.

** (xfce4-power-manager:22543): WARNING **: 12:54:46.268: Error: The connection is closed

** (xfce4-power-manager:22543): WARNING **: 12:54:46.269: Error: The connection is closed
```